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OVEN WITH ADJUSTABLE VOLUME PROCESSING CHAMBER

Abstract

An oven with a processing chamber having an adjustable enclosed volume that is configured to support a substrate and includes a lamp assembly configured to heat the substrate. The oven also includes a sealing door that bounds one end of the processing chamber. The sealing door is configured to be moved from a first position to a second position to change the enclosed volume of the processing chamber from a first volume to a second volume.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an oven with a processing chamber having an adjustable enclosed volume.

BACKGROUND

[0002] The electronics industry has continued to engineer greater functionality into smaller electronic devices at lower cost. This drive towards smaller and less expensive electronic devices has driven the development of advanced semiconductor packaging technologies. For example, wire bonds conventionally used to attach a die to a substrate, have been replaced by solder bumps. Flip chip, also known as controlled collapse chip connection (abbreviated as C4), is one such method of connecting semiconductor dies (or IC chips) to a substrate with solder bumps deposited on the I/O (input/output) pads of the dies on the top side of the wafer before the wafer is diced into individual dies. After the wafer is diced into individual dies, the die is flipped so that its top side faces down and aligned so that its pads align with matching pads on the substrate. The solder is reflowed to complete the interconnect.

[0003] Conventionally, solder reflow is accomplished by passing the assembly through a batch reflow oven in which the assembly passes through different zones of the oven on a conveyor belt. The zones heat and cool the wafers and introduce chemical vapor to the wafers at atmospheric pressure. Atmospheric pressure may provide a competing force to the chemical vapor pressure and hinder the vapor from reaching all regions of the solder bumps. Moreover, conventional reflow systems may require a relatively large amount of chemicals and, as a result, increase the cost of ownership and operation. The possibility of chamber contamination may also be high resulting in more frequent cleaning and maintenance requirements. The configurable ovens of the current disclosure may alleviate one or more of the above-described issues.

SUMMARY

[0004] Several embodiments of a processing oven having an adjustable enclosed volume is disclosed. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only. As such, the scope of the disclosure is not limited solely to the disclosed embodiments. Instead, it is intended to cover such alternatives, modifications and equivalents within the spirit and scope of the disclosed embodiments. Persons skilled in the art would understand how various changes, substitutions and alterations can be made to the disclosed embodiments without departing from the spirit and scope of the disclosure.

[0005] In one embodiment, an oven is disclosed. The oven may include a processing chamber having an adjustable enclosed volume. The processing chamber may include a spindle configured to support a substrate. The oven may also include a lamp assembly configured to heat the substrate supported on the spindle. The lamp assembly may include a plurality of spaced-apart lamps. The oven may further include a sealing door that bounds one end of the processing chamber. The sealing door may be configured to be moved from a first position to a second position to change the enclosed volume of the processing chamber from a first volume to a second volume.

[0006] Various embodiments of the disclosed oven may alternatively or additionally include one or more of the following features: the second volume may be between about 25%-75% of the first volume; the second volume may be less than 50% of the first volume; the sealing door may bound a bottom end of the processing chamber and the lamp assembly may be positioned at a top end of the processing chamber; the spindle may be configured to rotate with the substrate about a central axis of the processing chamber; the substrate may be positioned closer to the lamp assembly when the sealing door is located in the second position than when the sealing door is located in the first position; the oven may further include a chemical delivery tube positioned in processing chamber and the chemical delivery tube may include multiple spaced-apart ports configured to discharge a gas into the processing chamber; a distance of the chemical delivery tube from a top end of the processing chamber may be substantially same as the distance of the substrate from the top end

when the sealing door is located in the second position; the chemical delivery tube may be positioned radially outwards of an outer periphery of the substrate and radially inwards of a side wall of the processing chamber; the chemical delivery tube may have an arc-shape and subtend a sector angle between about 70-1200 about a central axis of the processing chamber; the sector angle may be about 900; the processing chamber may include a lid and the lamp assembly may be disposed on an underside of the lid; a side wall of the processing chamber may include a substrate-inlet port configured to direct the substrate into the enclosed volume of the processing chamber; the substrate-inlet port may be positioned above the sealing door when the sealing door is located in the first position, and may be positioned below the sealing door when the sealing door is located in the second position.

[0007] In another embodiment, an oven is disclosed. The oven may include a processing chamber having an adjustable enclosed volume, a lamp assembly at a top end of the processing chamber, and a sealing door at a bottom end of the processing chamber. The processing chamber may be configured to support a substrate and the lamp assembly may include a plurality of spaced-apart lamps configured to heat the substrate. The sealing door may be configured to be moved from a first position to a second position to decrease the enclosed volume of the processing chamber from a first volume to a second volume. The second volume may be between about 25%-75% of the first volume.

[0008] In a further embodiment, a method of processing a substrate in an oven is disclosed. The method may include loading the substrate on a spindle positioned in a processing chamber of the oven. The processing chamber may have an adjustable enclosed volume bounded by a sealing door located at one end of the processing chamber. The method may also include moving the sealing door from a first position to a second position to change the enclosed volume of the processing chamber from a first volume to a second volume. The method may further include activating a lamp assembly of the oven to heat the substrate, wherein the lamp assembly includes a plurality of spaced-apart lamps.

[0009] Various embodiments of the disclosed method may alternatively or additionally include one or more of the following features: the second volume may be between about 25%-75% of the first volume; the second volume may be less than 50% of the first volume; the sealing door may bound a bottom end of the processing chamber and the lamp assembly may be positioned at a top end of the processing chamber; rotating the spindle with the substrate about a central axis of the processing chamber; the substrate may be positioned closer to the lamp assembly when the sealing door is located in the second position than when the sealing door is located in the first position; discharging a gas into the processing chamber via multiple spaced-apart ports of a chemical delivery tube positioned in processing chamber; a distance of the chemical delivery tube from a top end of the processing chamber may be substantially same as the distance of the substrate from the top end when the sealing door is located in the second position; the chemical delivery tube may be positioned radially outwards of an outer periphery of the substrate and radially inwards of a side wall of the processing chamber; the chemical delivery tube may have an arc-shape and subtend a sector angle between about 70-1200 about a central axis of the processing chamber; the sector angle may be about 900; the processing chamber may include a lid and the lamp assembly may be disposed on an underside of the lid; a side wall of the processing chamber may include a substrate-inlet port configured to direct the substrate into the enclosed volume of the processing chamber; the substrate-inlet port may be positioned above the sealing door when the sealing door is located in the first position, and may be positioned below the sealing door when the sealing door is located in the second position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate exemplary embodiments and, together with the description, are used to explain the disclosed principles. In these drawings, where appropriate, reference numerals illustrating like structures, components, materials, and/or elements in different figures are labeled similarly. It is understood that various combinations of the structures, components, and/or elements, other than those specifically shown, are contemplated and are within the scope of the present disclosure.

[0011] For simplicity and clarity of illustration, the figures depict the general structure of the various described embodiments. Details of well-known components or features may be omitted to avoid obscuring other features, since these omitted features are well-known to those of ordinary skill in the art. Further, elements in the figures are not necessarily drawn to scale. The dimensions of some features may be exaggerated relative to other features to improve understanding of the exemplary embodiments. One skilled in the art would appreciate that the features in the figures are not necessarily drawn to scale and, unless indicated otherwise, should not be viewed as representing proportional relationships between different features in a figure. Additionally, even if it is not specifically mentioned, aspects described with reference to one embodiment or figure may also be applicable to, and may be used with, other embodiments or figures.

[0012] FIGS. **1**A and **1**B illustrate an exemplary oven of the current disclosure;

[0013] FIG. **2** illustrates an exemplary substrate configured to processed in the oven of FIG. **1**A;

[0014] FIG. **3** illustrates an exemplary temperature profile that may be used to process the substrate of FIG. **2**;

[0015] FIG. **4** illustrates an exemplary spindle of the oven of FIG. **1**A;

[0016] FIGS. **5**A and **5**B are schematic side views of the processing chamber of the oven of FIG. **1**A;

[0017] FIG. **6** illustrates a perspective view of an exemplary oven with its lid open;

[0018] FIG. **7** is a flow chart of an exemplary process that may be carried out in the oven of FIG. **1**A;

[0019] FIG. **8**A is perspective view showing a portion of an exemplary processing chamber of the oven of FIG. **1**A;

[0020] FIG. **8**B is a schematic top view of an exemplary processing chamber of the oven of FIG. **1**A;

[0021] FIGS. **9**A-**9**B are schematic diagrams illustrating some exemplary ovens of the current disclosure; and

[0022] FIG. **10** is a schematic diagram illustrating another exemplary oven of the current disclosure.

DETAILED DESCRIPTION

[0023] All relative terms such as “about,” “substantially,” “approximately,” etc., indicate a possible variation of ±10% (unless noted otherwise or another variation is specified). For example, a feature disclosed as being about “t” units long (wide, thick, etc.) may vary in length from (t−0.1t) to (t+0.1t) units. Similarly, a temperature within a range of about 100-150° C. can be any temperature between (100−10%) and (150+10%). In some cases, the specification also provides context to some of the relative terms used. Similarly, term “generally” implies a degree of approximation or similarity rather than an exact replication. For example, a structure described as being substantially circular or generally circular, it means that the structure shares similarities with the geometric characteristics of a circle but may not precisely match its form. For example, its shape may deviate slightly (e.g., 10% variation in diameter or curvature at different locations, etc.) from being perfectly circular. Further, a range described as varying from, or between, 5 to 10 (5-10), includes the endpoints (i.e., 5 and 10).

[0024] Unless otherwise defined, all terms of art, notations, and other scientific terms or terminology used herein have the same meaning as is commonly understood by one of ordinary skill in the art to which this disclosure belongs. Some of the components, structures, and/or processes described or referenced herein are well understood and commonly employed using conventional methodology by those skilled in the art. Therefore, these components, structures, and processes will not be described in detail. All patents, applications, published applications and other publications referred to herein as being incorporated by reference are incorporated by reference in their entirety. If a definition or description set forth in this disclosure is contrary to, or otherwise inconsistent with, a definition and/or description in these references, the definition and/or description set forth in this disclosure controls over those in the references that are incorporated by reference. None of the references described or referenced herein is admitted as prior art to the current disclosure.

[0025] FIG. **1**A is a side view and FIG. **1**B is a perspective view of an exemplary processing oven **100** of the current disclosure. In the discussion below, processing oven **100** will be described with reference to an exemplary use case, for example, as a solder reflow oven. However, this use case is only exemplary, and oven **100** can be used for any high temperature (e.g., curing polymers, etc.) application. Oven **100** includes a processing chamber **40** configured to receive one or more substrates (e.g., wafer, organic/ceramic substrates, semiconductor packages, printed circuit board (PCB), etc.) and subject the substrates to a high temperature process. In the discussion below, a semiconductor wafer with solder on its top surface (see FIG. **2**) will be described as the substate that is subject to an exemplary solder reflow process in oven **100**. In general, any size (e.g., 200 mm, 300 mm, etc.) of semiconductor wafer may be received in chamber **40**. FIG. **2** illustrates an exemplary semiconductor wafer **10** that may be received in chamber **40** and subject to a reflow process. As would be recognized by a person skilled in the art, after integrated circuit processing, wafer **10** includes multiple dies (or IC chips), and solder material may be deposited on the I/O pads of the multiple dies in wafer **10**. Prior to dicing the wafer **10** into individual dies, wafer **10** may be subject to a reflow process in oven **100** to form solder bumps **12** thereon (called wafer bumping). Any type of solder material (lead-free solder, lead-tin solder, etc.) may be deposited on wafer **10** and the wafer **10** may be subject to any type of reflow process or reflow profile. Persons having ordinary skill in the art would recognize that reflow process that wafer **10** is subjected to depends on the type of solder material deposited on wafer **10**. Suitable reflow profiles for different types of solder materials are available in the published literature and may be obtained from solder vendors. FIG. **3** illustrates an exemplary reflow profile that may be applied to wafer **10** in chamber **40**. In FIG. **3**, the X-axis shows time in seconds, and the Y-axis shows temperature in degrees Celsius (° C.).

[0026] FIGS. **5**A and **5**B are schematic illustrations of an exemplary chamber **40** of oven **100**. A robotic manipulator or arm (not shown) may insert wafer **10** into chamber **40** through an inlet port **42** (see FIG. **1**B). Wafer **10** may be disposed on a spindle **44** within chamber **40**. FIG. **4** illustrates an exemplary spindle **44** separated from oven **100**. With specific reference to FIG. **4**, spindle **44** may include a plurality of arms **46**A, **46**B, **46**C (e.g., **3** arms in the embodiment of FIG. **4**) that extend radially outward from a central axis **50** of chamber **40**. Each arm **46**A, **46**B, **46**C of spindle **44** includes nub **48**A, **48**B, **48**C that projects upwards. Wafer **10** may rest on nubs **48**A, **48**B, **48**C of spindle **44**. Oven **100** also includes a motor assembly **20** (see FIGS. **1**A-**1**B) that may be configured to support and control the movement of wafer **10** in chamber **40**. In some embodiments, motor assembly **20** may include one or motors configured to rotate spindle **44** (and wafer **10** supported thereon) in chamber **40** about its central axis **50**. Spindle **44** may be rotated at any speed. In some embodiments, the rotational speed of spindle **44** may vary between about 0-20 RPM. Motor assembly **20** may also include one or more motors to raise and lower the rotating spindle **44** and/or other components (e.g., sealable door **48**) of oven **100** within chamber **40** along axis **50**.

[0027] In some embodiments, chamber **40** may have a substantially cylindrical configuration (see FIG. **1**B). However, a substantially cylindrical shape is not a requirement. In general, chamber **40** can have any shape (see, e.g., FIG. **10**). In some embodiments, the chamber **40** may be substantially square, substantially rectangular, or have another shape. With specific reference to FIGS. **5**A-**5**B, chamber **40** may have an enclosed volume bounded (or defined) by a top wall or lid **46**, a bottom wall or sealing door **48**, and a side wall **52** extending between lid **46** and door **48**. In some embodiments, lid **46** may be hinged on side wall **52** and configured to rotate about the hinge between an open position and a closed position. When lid **46** is closed, an O-ring **64** or another sealing device between lid **46** and side wall **52** may seal the enclosed volume of chamber **40** from the outside atmosphere. Side wall **52** of chamber **40** defines a slit or an opening that forms inlet port **42** (see FIG. **1**B). In some embodiments, a valve, flap, or another closing mechanism may seal the opening of inlet port **42** such that the enclosed volume of chamber **40** can be pumped down to a low pressure.

[0028] Oven **100** may include a lamp assembly **80** including a plurality of lamps **82** configured to heat wafer **10** positioned in chamber **40**. In some embodiments, lamp assembly **80** may be positioned within chamber **40** and configured to heat the top surface of wafer **10** positioned in chamber **40**. In some embodiments, lamps **82** of lamp assembly **80** may be arranged on, or coupled to, the underside of lid **46** such that the lamp assembly swivels about the hinge along with the lid. When activated, the lamps **82** of lamp assembly **80** heats the top surface of wafer **10** in chamber **40**. In general, lamp assembly **80** may include any number and any type of lamps **82**. In some embodiments, lamp assembly **80** may include 4-10 infrared (IR) lamps having a power between about 1-10 kW for each lamp, or between about 1.5-3 kW (or about 2 kW) for each lamp **82**. In some embodiments, lamp assembly **80** may include seven halogen lamps **82**. The lamps **82** may be arranged (e.g., spaced apart) such that they evenly heat the top surface of wafer **10** in chamber **40**. Details of the lamp assembly and oven **100** in some exemplary embodiments of the current disclosure are described in U.S. Pat. No. 11,296,049 incorporated by reference in its entirety herein.

[0029] The size (e.g., width, diameter, etc.) of chamber **40** depends upon the application, for example, the size of wafer **10** that will be processed in chamber **40**. In some embodiments, a chamber **40** configured to process 300 mm wafers may have a diameter of about 450 mm. However, this size is only exemplary, and chamber **40** may have any size. Ovens of the current disclosure are configured such that the height of chamber **40** (and consequently the enclosed volume of the chamber) may be adjusted (e.g., varied or changed). In some embodiments, the height of chamber **40** (and consequently its volume) may be changed by raising and lowering sealing door **48** that forms the bottom wall of chamber **40**. FIG. **5**A illustrates a configuration of oven **100** with sealing door **48** disposed at its lower position (or a first position) and FIG. **5**B illustrates a configuration of oven **100** with sealing door **48** disposed at its upper position (or a second position). The sealing door **48** may be moved between its lower and upper positions (i.e., first and second positions) by a motor of motor assembly **80**. As illustrated in FIGS. **5**A and **5**B, the height and enclosed volume of chamber **40** is smaller when sealing door **48** is located in its upper position and larger when sealing door **48** is located in its lower position. In some embodiments, the enclosed volume of chamber **40** when sealing door **48** is located in its upper position may be between about 25%-75% of its enclosed volume when sealing door **48** is located in its lower position. In some embodiments, the enclosed volume of chamber **40** when sealing door **48** is located in its upper position may be between about 40%-60% of its enclosed volume when sealing door **48** is located in its lower position. In some embodiments, the enclosed volume of chamber **40** may be more than 50% smaller when sealing door **48** is located in its upper position than when sealing door **48** is located in its lower position.

[0030] The values of chamber height when sealing door **48** is positioned in the upper and lower positions depend on the application (e.g., process carried out, number of samples being treated, etc.). In some exemplary embodiments, the height “H.sub.1” of chamber **40** (e.g., the distance between the top surface of sealing door **48** and the bottom surface of lamp assembly **80**) when sealing door **48** is in the lower position is between about 75-150 mm and the height “H.sub.2” of chamber **40** when sealing door **48** is in the upper position is between about 20-50 mm. It should be noted that these heights are merely exemplary, and in general, these heights may have any values. In some embodiments, the distance between the top surface of wafer **10** seated on spindle **44** and the underside of lamp assembly **10** may be between about 10-50 mm when sealing door **48** is at its upper position (see FIG. **5**B), and between about 75-150 mm when sealing door **48** is at its lower position (see FIG. **5**A). In some embodiments, this distance may be between about 20-30 mm when sealing door **48** is at its upper position, and about 90-120 mm when sealing door **48** is at its lower position.

[0031] In some embodiments, in addition to sealing door **48**, spindle **44** may also be configured to move vertically up and down along central axis **50**. A motor of motor assembly **20** may be configured to move the rotating spindle **44** (along wafer **10** supported thereon) vertically along central axis **50**. For example, when sealing door **48** is located at its lower or upper position, spindle **44** may move wafer **10** towards or away from lamp assembly **80**. In other words, in some embodiments of oven **100**, both sealing door **48** and spindle **44** may be individually configured to move wafer **10** towards and away from lamp assembly **80**. However, this is only exemplary, and in some embodiments of oven **100**, spindle **44** may not be configured to move along central axis **50** independent from sealable door **48**. In such embodiments, only sealing door **48** may be configured to move wafer **10** towards and away from lamp assembly **80**, and spindle **44** may move up and down along with sealing door **48**. When sealing door **48** is located at its upper position, wafer **10** (on spindle **44**) may be positioned closer to lamp assembly **80** than when sealing door **48** is positioned at its lower position.

[0032] It should be noted that although sealing door **48** is described as having two positions (e.g., a lower and an upper position) to adjust the enclosed volume of chamber **40** between two values (e.g., a lower volume and a higher volume), this is only exemplary. In some embodiments, sealing door **48** may be moved and fixed at more than two positions (e.g., three, four, five, etc.) to create corresponding changes in the enclosed volume of chamber **40**. In general, sealing door **48** may be moved and fixed (or located) at multiple positions vertically spaced apart from each other to produce corresponding changes in the enclosed volume of chamber **40**. For example, in addition to the lower and upper (or first and second) positions illustrated in FIGS. **5**A and **5**B, sealing door **48** may also be moved and fixed at one or more locations between these two positions to result in chamber **40** having multiple enclosed volumes.

[0033] Sealing door **48** may be moved vertically up and down to selectively define multiple enclosed volumes (e.g., a smaller volume and a larger volume in the embodiment of FIGS. **5**A and **5**B) of chamber **40** in any manner. In some embodiments, as illustrated in FIGS. **5**A and **5**B, side wall **52** of chamber **40** may include an upper portion **52**A and a lower portion **52**B that is larger in size (e.g., diameter, width, etc.) than upper portion **52**A. The top end of the upper portion **52**A may be covered by lid **46** and the bottom end of the upper portion **52**A may be connected to the top end of the lower portion **52**B at a ledge **60**. And the bottom end of the lower portion **52**B may be covered by a bottom wall **58**. A housing **56** may be defined by lid **46** and bottom wall **58** positioned on opposite sides of side wall **52**. Sealing door **48** may be configured to move vertically up and down in the lower portion **52**B of side wall **52** between ledge **60** and bottom wall **58**. In its lower position, sealing door **48** may be positioned proximate (or mate with) bottom wall **58** to define the larger enclosed volume of chamber **40**. When sealing door **48** is moved to its upper position, it may mate with ledge **60** to reduce the enclosed volume of chamber **40**. An O-ring **62** positioned between the bottom surface of ledge **60** and the top surface of sealing door **48** may seal the reduced enclosed volume of chamber **40** from the outside atmosphere.

[0034] In some embodiments, inlet port **42** through which wafer **10** is inserted into chamber **40** may be located on housing **56** in lower portion **52**B of side wall **52** such that, when sealing door **48** is located at its upper position, the inlet port is positioned below the sealing door (see FIG. **5**B), and therefore, fluidly decoupled from the enclosed volume of chamber **40**. In contrast, when sealing door **48** is located at its lower position (see FIG. **5**A), the inlet port is positioned above the sealing door and therefore fluidly coupled to the enclosed volume of chamber **40**. It should be noted that the above-described configuration is only exemplary, and ovens of the current disclosure may have other configurations.

[0035] FIG. **6** illustrates a perspective view of an exemplary oven **100** with its lid **46** in an open position. Housing **56** of oven **100** may include one or more gas ports **72**. Although only one gas port **72** is shown in FIG. **6**, in some embodiments, housing **56** may include multiple gas ports **72**. Process gas or an inert gas (e.g., nitrogen gas) may be selectively directed into chamber **40** through these gas port **72**. In some embodiments, vacuum pump may be coupled to one or more gas port **72** to generate a vacuum in chamber **40**. In some embodiments, side wall **52** of chamber **40** may include multiple openings (e.g., gas inlet ports **74**) configured to direct process gas and/or inert gas into (or exhaust the gas from) chamber **40**. In the embodiment of FIG. **6**, lamps **82** of lamp assembly **80** are positioned on the underside of lid **46**. In some embodiments, lamps **82** may be arranged closer together at the edges (of lid **46**) than at the center. As a result of edge effects, wafer **10** in chamber **40** may cool down quicker at the edges than at the center. A smaller relative spacing of lamps **82** at the edges than at the center may assist in achieving an even temperature distribution on wafer **10**.

[0036] In some embodiments, lamps **82** may be controlled by control system **200** (schematically illustrated in FIG. **1**A) of oven **100** to heat wafer **10** in chamber **40** to a desired temperature at a selected temperature ramp rate. For example, a different number of lamps **82** may be activated to increase or decrease the temperature and/or the ramp rate. Alternatively, or additionally, in some embodiments, the power of lamps **82** may be varied to vary the temperature and/or the ramp rate. In some embodiments, control system **200** may control the power to lamps **82** based on the detected temperature in chamber **40** (e.g., using a feedback loop). For example, when thermocouples (or pyrometers or another temperature detection sensor) in chamber **40** indicate that the temperature of wafer **10** is below a desired value, the control system **200** may increase the power to lamps **82**. In some embodiments, when the thermocouples indicate that the temperature variation in wafer **10** is above a threshold value (e.g., temperature at the edges of the wafer is below the temperature at the center, etc.), control system **200** may vary the power to different lamps **82** in lamp assembly **80**. Although not visible in FIG. **6**, in some embodiments, a transparent window (e.g., made of a material that is transparent to the wavelength of light emanating from lamps **82**) may be provided at the bottom of lamp assembly **80** to isolate and protect lamps **82** from the environment within chamber **40**. In such embodiments, lamps **82** heat the wafer in chamber **40** through the transparent window. In some embodiments, the window may be made of, for example, quartz, glass, etc.

[0037] An exemplary process **700** using oven **100** will now be described with reference to FIG. **7**. In step **710**, a wafer (e.g., wafer **10** of FIG. **2**) may be loaded into chamber **40** of oven **100**. When the wafer is loaded, sealing door **48** of oven **10** may be located at its lower position (as illustrated in FIG. **5**A) such that chamber **40** has a larger enclosed volume and spindle **44** is accessible via inlet port **42**. Wafer **10** may be inserted into chamber **40** through inlet port **42** and positioned on nubs **48**A-**48**D of spindle **44** (see FIG. **4**). In some embodiments, in step **710**, inlet port **42** is opened, wafer **10** is inserted into the chamber through the open inlet port and positioned on spindle **44**, and inlet port **42** is then closed. In step **720**, chamber **40** with the loaded wafer may be pumped down in pressure. In general, in this step, the pressure in chamber **40** may be pumped down to any value of pressure below atmospheric pressure. Chamber **40** may be pumped down to a sub-atmospheric pressure using vacuum pump **140** (see FIGS. **9**A-**9**B). In some embodiments, chamber **40** may be pumped down to a pressure between about 100 milliTorr-100 Torr in this step.

[0038] In step **730**, the sealing door **48** may be moved (e.g., translated) to its upper position to reduce the enclosed volume of chamber **40** (see FIG. **5**B). The sealing door **48** may be moved to its upper position by activating a motor of motor assembly **20**. When the sealing door **48** is positioned in its upper position, the enclosed volume of chamber **40** is reduced and decoupled from inlet port **42**. In step **740**, the reduced enclosed volume of the chamber **40** is pumped down to a lower pressure (e.g., than in step **720**). Chamber **40** may be pumped down to any pressure between about 100 milliTorr-100 Torr in this step. For example, if the chamber is pumped down to a pressure of 100 Torr in step **720**, in step **740**, the reduced volume of the chamber may be pumped down to any pressure lower than 100 Torr (e.g., 50 Torr). In some embodiments, in step **740**, the chamber may be pumped down to a pressure between about 1-10 Torr.

[0039] In step **750**, any desired thermal processing of wafer **10** may be performed. Any high temperature process may be carried out in step **750** with the heat for the process provided by lamps **82** of lamp assembly **80**. For example, if the thermal process in step **750** involves curing a polymer coating on the top surface of wafer **10**, lamps **82** of lamp assembly **80** may be activated to heat wafer **10** to the desired curing temperature (e.g., at the desired ramp rate) for curing the polymer. As another example, if the thermal process in step **750** involves reflowing solder bumps **12** (see FIG. **2**) on wafer **10**, control system **200** of oven **100** may control lamp assembly **80** to subject the solder bumps **12** to the reflow profile (e.g., the exemplary reflow profile of FIG. **3**) of the solder. To subject wafer **10** to, for example, the solder reflow profile of FIG. **3**, control system **200** may activate lamps **82** and/or control the power to lamps **82** to heat wafer **10** in accordance with this reflow profile based on signals from thermocouples or pyrometers in chamber **40**. In some embodiments, in step **750**, spindle **44** may be rotate wafer **10** to evenly heat the top surface of wafer**10**.

[0040] In some embodiments, a chemical vapor (e.g., formic acid vapor) may also be directed into the reduced enclosed volume of chamber **40** during step **750**. For example, the chemical vapor may be directed into the enclosed volume during selected stages of the reflow profile. Rotating wafer **10** (by rotating spindle **44**) may ensure that the entire top surface of wafer is evenly exposed and coated with the chemical vapor. As will be described in more detail later, the chemical vapor may be delivered into chamber **40** via a chemical delivery tube **90**. In some embodiments, inert gas (e.g., nitrogen) may also be admitted into chamber **40** during some stages of the thermal processing. Inert gas may be admitted into chamber, for example, via gas inlet ports **74** on side wall **52** of chamber **40**. In some embodiments, during step **750**, chamber **40** may be evacuated (e.g., using vacuum pump **140** of FIGS. **9**A-**9**B) during some stages of the thermal processing to remove excess vapor from the enclosed volume of chamber **40**.

[0041] In step **760**, the enclosed volume of chamber **40** may be vented to atmosphere. In some embodiments, inert gas (e.g., nitrogen) may also be directed into the enclosed volume of chamber **40** during this step, for example, to cool wafer **10**. In step **770**, the sealing door **48** may be moved to its lower position such that the enclosed volume of chamber **40** is increased and wafer **10** is accessible via inlet port **42**. In some embodiments, the inert gas may continue to flow (e.g., via ports **74**) into the increased enclosed volume of chamber **40** during this step to cool the wafer. After wafer **10** is cooled to the desired temperature, the wafer may be removed from chamber **40** via the inlet port **42** in step **780**.

[0042] As explained previously, during some thermal processes carried out in step **750**, a chemical vapor (e.g., formic acid vapor) or another process gas (e.g., a chemical in gaseous form) may injected into the enclosed volume of chamber **40** via a chemical delivery tube **90**. FIG. **8**A illustrates a perspective view of a portion of an exemplary oven **100** with a chemical delivery tube **90** positioned in chamber **40**. In general, chemical delivery tube **90** may have a shape configured to discharge the chemical vapor or gas substantially uniformly on wafer **10**. Chemical delivery tube **90** may have multiple nozzles or ports **92** to dispense the gas into chamber **40**. In some embodiments, the multiple ports **92** may be arranged to dispense the gas in a generally upward direction (e.g., towards lid **46**). In some embodiments, ports **92** may be arranged to direct the gas generally sideways towards the side wall **52** of chamber **40** away from wafer **10** positioned in chamber **40**. The gas exiting these ports **92** may reflect off the side wall and flow over wafer **10**. In some embodiments, the multiple ports **92** may be arranged to dispense the gas sideways towards wafer **10**. Any number (2-20) of ports **92** may be provided on tube **90**.

[0043] FIG. **8**B is a schematic top view of chamber showing an exemplary chemical delivery tube **90** positioned radially outwards of the wafer positioned in the chamber. In some embodiments, as illustrated in FIG. **8**A, chemical delivery tube **90** may be shaped like an arc extending circumferentially along a portion of the side wall of chamber **40**. Multiple nozzles **92** may be dispersed along the length of the arc-shaped tube **90**. As used herein, arc-shaped tube refers to a tube or pipe that has a curved shape resembling the form of an arc. In some embodiments, as illustrated in FIG. **8**B, the arc-shaped chemical delivery tube **90** may be positioned radially outwards of the wafer **10**. In other words, chemical delivery tube **90** may be an arc-shaped tube with gas-discharge ports that extends circumferentially between the perimeter of wafer **10** and the side wall **52** of chamber **40**. As illustrated in FIG. **8**B, the arc-shaped chemical delivery tube **90** may generally form a sector of a circle positioned radially between the perimeter of wafer **10** and the side wall **52** of chamber **40**. In other words, the arc-shaped tube's curved shape may generally follow the boundary of a sector (or portion) of a circle. That is, the shape of the arc-shaped tube shares similarities with the geometric characteristics of a circular sector but may not precisely match its form. The sector angle θ, which defines the arc length or extent of the arc-shaped tube **90**, may be any value. As used herein, and as illustrated in FIG. **8**B, sector angle θ is the angle between two lines originating from central axis **50** (or the center of wafer **10**) and extending to the two outer edges of the arc-shaped tube **90**. In some embodiments, the length of chemical delivery tube **90** may be such that the sector angle θ is between about 70-120°, or between about 80-100°. In some embodiments, the length of tube **90** may be such that the sector angle θ is about 90°.

[0044] In some embodiments, as illustrated in FIG. **5**B, the vertical position of chemical delivery tube **90** in chamber **40** may be such that, when sealing door **48** is positioned at its upper (or second) position, tube **90** is positioned adjacent to, or alongside, wafer **10** on spindle **44**. In some embodiments, the vertical position of tube **90** in chamber **40** may be substantially equal to the vertical position of wafer **10** in chamber **40** when sealing door **48** is positioned at its upper position. In other words, the distance between tube **90** and the underside of lamp assembly **80** (or lid **46**) may be substantially the same as the distance between the top surface of wafer **10** and the underside of lamp assembly **80** (or lid **46**) when sealing door **48** is positioned at its upper position. Dispersing chemical vapor into chamber **40** through ports **92** of an arc-shaped chemical delivery tube **90** positioned adjacent to the wafer may enable a laminar flow of the vapor over the wafer surface and assist in treating solder bumps in all areas of wafer **10** substantially uniformly.

[0045] In some embodiments, as illustrated in FIG. **8**B, chamber **40** may also include a chemical removal tube **94** with multiple exit ports **96** positioned diametrically across from chemical delivery tube **90**. Chemical removal tube **94** may also be an arc-shaped tube extending circumferentially between the perimeter of wafer **10** and the side wall **52** of chamber **40**. Similar to the arc-shaped chemical delivery tube **90**, the chemical removal tube **94** may also generally form a sector of a circle having any sector angle. The vertical position of removal tube **94** in chamber **40** may be substantially equal to the vertical position of delivery tube **90** in chamber **40**. A chemical vapor discharged or admitted into chamber **40** through ports **92** of chemical delivery tube **90** may treat the surface of wafer **10** during step **750** (of process **700** of FIG. **7**) and be evacuated or removed from chamber **40** through ports **96** of chemical removal tube **94**. In some embodiments, chemical removal tube **94** may be eliminated and the chemical vapor may be removed from chamber **40** through ports (e.g., ports **74**, see FIG. **6**) positioned on side wall **52** of chamber **40**.

[0046] It should be noted that, although chemical delivery tube **90** is described as being used to deliver a chemical vapor into chamber **40**, this is only exemplary. In general, chemical delivery tube **90** may be used to discharge any gas (or gaseous chemical) into the enclosed volume of chamber **40**. The type of gas or vapor discharged into chamber **40** depends upon the type of thermal processing being performed in step **750** (in method **700** of FIG. **7**). As illustrated in FIG. **9**A, chemical delivery tube **90** may be fluidly connected to a gas (or chemical vapor) supply **110** that contains (e.g., a tank) or generates (e.g., a device such as a bubbler that produces) the gas or vapor being directed into chamber **40**. The gas or vapor from gas supply **110** may be directed or admitted into chamber through the ports **92** of chemical delivery tube **90**.

[0047] As also illustrated in FIG. **9**A, ports **96** of chemical removal tube **94** (or the ports **74** (see FIG. **6**) on the side wall **52** of chamber **40** that removes the vapor or gas from chamber **40**) may be connected to a vacuum pump **140**. Vacuum pump **140** may be activated (e.g., by control system **200**) to remove the gas or vapor from chamber **40** and pump down chamber **40** to a sub-atmospheric pressure. In some embodiments, as illustrated in FIG. **9**A, chamber **40** may be fluidly connected to vacuum pump **140** through an isolation valve **120** and a throttle valve **130**. Isolation valve **120** may allow controlled start and stop of the vacuum process. It may be closed to isolate chamber **40** from vacuum pump **140** to enable maintenance, repairs, or adjustments without affecting the entire system. When vacuum pump **140** is not in operation, isolation valve **120** may prevent backflow of gases or contaminants into chamber **40** and assist in maintaining the desired vacuum level and cleanliness of chamber **40**. Throttle valve **130** may allow for controlled regulation of gas flow into vacuum pump **140** and assist in regulating pumping speed and optimizing the vacuum level within chamber **40**. Control system **200** may adjust the opening of throttle valve **130** to control the pressure within chamber **40**.

[0048] In some embodiments, as illustrated in FIG. **9**B, the gas removal connection between chamber **40** and vacuum pump **140** may include a bypass line **150** with a second isolation valve **160** and a regulator **170**. Regulator **170** may control the flow of gas through bypass line **150** and allow for fine-tuning and controlling the flow of gas independent of isolation valve **120**. The bypass line **150** may provide flexibility in adjusting the gas flow independently of the main line and may be useful in processes where precise control over the gas flow rate is required. The bypass line **150** may also assist in preventing overloading of vacuum pump **140** by providing an alternative path for gas to flow. The isolation valve**160** in the bypass line **150** may allow for controlled activation or deactivation of the bypass line **150**. It can be closed to force gas flow through the main line or opened to allow gas to bypass the main isolation valve **120**. In other words, the bypass line **150** with a regulator **170** and isolation valve **160** may add a layer of control and flexibility to the vacuum system, for example, by allowing independent adjustment of gas flow, by providing an alternative path to protect the vacuum pump, and by facilitating maintenance or adjustments without disrupting the main vacuum process.

[0049] It should be noted that an arc-shape of chemical delivery tube **90** and chemical removal tube **94** is not a requirement. In general, the shape of chemical delivery tube **90** and chemical removal tube **94** may depend on the cross-sectional shape of chamber **40**. FIG. **10** is a schematic illustration of a top view of an exemplary chamber **40′** in some embodiments of oven **100**. As illustrated in FIG. **10**, in some embodiments, chamber **40′** may have a square or rectangular cross-sectional shape and may be used to process a rectangular or square-shaped substrate **10′**. In such embodiments, chemical delivery tube **90′** may be a linear tube that extends along a side (e.g., width) of substrate **10′**. Multiple ports **92** may be provided on the linear-shaped delivery tube **90′** to discharge chemical vapor into chamber **40′**. Chamber **40′** may also include a linear-shaped chemical removal tube **94′** positioned opposite chemical delivery tube **90′**. Multiple spaced-apart ports **96** may be provided on chemical removal tube **94′** to remove chemical vapor from chamber **40′**. The vertical position of tubes **90′** and **94′** may be aligned with the vertical position of substrate **10′** as discussed above with reference to tubes **90** and **94**. In some embodiments, chemical removal tube **94′** may be eliminated and ports **74** (see FIG. **6**) on the side walls of chamber **40′** may be used to remove the chemical vapor from the chamber **40′**.

[0050] Thus, in ovens of the current disclosure, the enclosed volume of the processing chamber may be varied and selected to be one of multiple values. In other words, the processing chamber is designed in such a way that its internal space or enclosed volume can be adjusted, and users have the flexibility to choose from different possible enclosed volume settings. The adjustment of enclosed volume may be made by users, operators, or automated systems (e.g., control system **200**). This ability to select different enclosed volumes allows for the customization and adaptation of the processing chamber based on specific needs and requirements. For example, this allows the chamber to accommodate and process various numbers and/or sizes of substrates efficiently. As another example, adjusting the enclosed volume to match the task can contribute to resource efficiency since a smaller volume may require less energy (e.g., to heat) and/or resources (e.g., chemicals) compared to a larger one. The configuration of the chemical delivery tube may also increase process efficiency by uniformly distributing and treating all areas of the processed substrate(s).

[0051] The above-described embodiments of ovens **100** and process **700** are only exemplary. Many variations are possible. As a person skilled in the art would recognize, the steps of process **700** need not be performed in the order illustrated in FIG. **7**. Some steps may be omitted and/or some steps added in other exemplary methods. For example, in some embodiments, lamps **82** may be activated before loading wafer into the chamber (i.e., step **710**) or after moving sealing door to its upper position (step **730**), etc. Persons of ordinary skill in the art would recognize and understand these possible variations to be within the scope of the present disclosure. Furthermore, although oven **100** is described in conjunction with a solder reflow process, this is only exemplary. A person of ordinary skill in the art would recognize that the oven can be used for any high temperature process on any type of substrate (e.g., wafer, organic/ceramic substrates, semiconductor packages, printed circuit board (PCB), etc.).

Claims

1. An oven, comprising: a processing chamber having an adjustable enclosed volume, the processing chamber including a spindle configured to support a substrate; a lamp assembly configured to heat the substrate supported on the spindle, wherein the lamp assembly includes a plurality of spaced-apart lamps; and a sealing door that bounds one end of the processing chamber, wherein the sealing door is configured to be moved from a first position to a second position to change the enclosed volume of the processing chamber from a first volume to a second volume.
2. The oven of claim 1, wherein the second volume is between about 25%-75% of the first volume.
3. The oven of claim 1, wherein the second volume is less than 50% of the first volume.
4. The oven of claim 1, wherein the sealing door bounds a bottom end of the processing chamber, and the lamp assembly is positioned at a top end of the processing chamber.
5. The oven of claim 1, wherein the spindle is configured to rotate with the substrate about a central axis of the processing chamber.
6. The oven of claim 1, wherein the substrate is positioned closer to the lamp assembly when the sealing door is located in the second position than when the sealing door is located in the first position.
7. The oven of claim 1, further including a chemical delivery tube positioned in processing chamber, wherein the chemical delivery tube includes multiple spaced-apart ports configured to discharge a gas into the processing chamber.
8. The oven of claim 7, wherein a distance of the chemical delivery tube from a top end of the processing chamber is substantially same as the distance of the substrate from the top end when the sealing door is located in the second position.
9. The oven of claim 7, wherein the chemical delivery tube is positioned radially outwards of an outer periphery of the substrate and radially inwards of a side wall of the processing chamber.
10. The oven of claim 7, wherein the chemical delivery tube has an arc-shape and subtends a sector angle between about 70-120° about a central axis of the processing chamber.
11. The oven of claim 10, wherein the sector angle is about 90°.

12. The oven of claim 1, wherein the processing chamber includes a lid, and the lamp assembly is disposed on an underside of the lid.
13. The oven of claim 1, wherein a side wall of the processing chamber includes a substrate-inlet port configured to direct the substrate into the enclosed volume of the processing chamber.
14. The oven of claim 13, wherein the substrate-inlet port is positioned above the sealing door when the sealing door is located in the first position, and the substrate-inlet port is positioned below the sealing door when the sealing door is located in the second position.
15. An oven, comprising: a processing chamber having an adjustable enclosed volume, the processing chamber configured to support a substrate; a lamp assembly positioned at a top end of the processing chamber, wherein the lamp assembly includes a plurality of spaced-apart lamps configured to heat the substrate; and a sealing door at a bottom end of the processing chamber, wherein the sealing door is configured to be moved from a first position to a second position to decrease the enclosed volume of the processing chamber from a first volume to a second volume, and wherein the second volume is between about 25%-75% of the first volume.
16. The oven of claim 15, wherein the substrate is positioned closer to the lamp assembly when the sealing door is located in the second position than when the sealing door is located in the first position.
17. The oven of claim 15, further including a chemical delivery tube positioned in processing chamber, wherein the chemical delivery tube includes multiple spaced-apart ports configured to discharge a gas into the processing chamber, and wherein a distance of the chemical delivery tube from the lamp assembly is substantially same as the distance of the substrate from the lamp assembly when the sealing door is located in the second position.
18. The oven of claim 17, wherein the chemical delivery tube has an arc-shape and is positioned radially outwards of an outer periphery of the substrate and radially inwards of a side wall of the processing chamber.
19. The oven of claim 18, wherein the chemical delivery tube subtends a sector angle between about 70-120° about a central axis of the processing chamber.
20. The oven of claim 15, wherein a side wall of the processing chamber includes a substrate-inlet port configured to direct the substrate into the enclosed volume of the processing chamber, wherein the substrate-inlet port is positioned above the sealing door when the sealing door is located in the first position, and the substrate-inlet port is positioned below the sealing door when the sealing door is located in the second position.